

Combined treatment with colchicine and Herba Taraxaci (*Tarazacum mongolicum* Hand.-Mazz.) attenuates Behcet's disease-like symptoms in mice and influences the expressions of cytokines

Seonghyang Sohn^{a,*}, Dongsik Bang^b, Seung Ihm Lee^a, Young Ae Kim^a, Eun-So Lee^c, Jee Yong Ha^d, Jang Hyun Kim^e, Suh Young Choi^f, Sungnack Lee^c

^aLaboratory of Cell Biology, Institute for Medical Sciences, Ajou University, Suwon 442-749, South Korea

^bDepartment of Dermatology, College of Medicine, Yonsei University, Seoul 120-751, South Korea

^cSchool of Medicine, Ajou University, South Korea

^dCollege of Oriental Medicine, Sangji University, South Korea

^eCollege of Oriental Medicine, Dongguk University, South Korea

^fHana Medical Foundation, South Korea

Received 29 August 2002; received in revised form 26 February 2003; accepted 27 February 2003

Abstract

Herbal medicine, Herba Taraxaci (*Tarazacum mongolicum* Hand.-Mazz.), was administered to mice with Behcet's disease (BD)-like symptoms induced by herpes simplex virus (HSV). BD is a chronic recurrent inflammatory disease. Herba Taraxaci (6 mg) was administered alone or in combination with 2 µg of colchicine to BD-like mice. Colchicine is a drug that is widely used as a medication for BD patients. The water extracts of Herba Taraxaci were administered orally once per day for 20 days. Eighty percent (8 of 10) of mice treated with Herba Taraxaci combined with colchicine showed improvement in mucocutaneous symptoms compared to 0% (0 of 10) of the nontreated group and 30% (3 of 10) treated with colchicine alone. Cytokine expression in spleen tissue collected from treated mice was analyzed by RT-PCR and FACS. Treatment with Herba Taraxaci induced IL-4 mRNA, and spleen from mice receiving the combined treatment (Herba Taraxaci and colchicines) showed an increased number of splenocytes staining with anti-IL-10 (46.8 ± 6.80) compared to Herba Taraxaci (35.4 ± 2.17) ($p < 0.05$) or colchicine alone (26.2 ± 4.47) ($p < 0.001$). These results suggest that the Herba Taraxaci may be an effective complementary agent in the treatment of BD.

© 2003 Elsevier Science B.V. All rights reserved.

Keywords: Herba Taraxaci; Colchicine; Inflammation; Behcet's disease; In vivo animal model; Cytokine

1. Introduction

Behcet's disease (BD) is a recurrent, chronic, and systemic disorder with mucocutaneous, ocular, arthritic, vascular, gastrointestinal, and central nervous

* Corresponding authors. S. Sohn is to be contacted at Tel.: +81-31-219-4510; fax: +82-31-219-4503. D. Bang is to be contacted at Tel.: +82-2-361-5734; fax: +82-2-361-9157.

E-mail addresses: sohnsh@madang.ajou.ac.kr (S. Sohn), dbang@yumc.yonsei.ac.kr (D. Bang).

system involvement. The etiology of BD has been linked to viral infection, autoimmune disease, streptococcal-related antigens, specific alleles of the human major histocompatibility complex, and hazardous chemicals [1–4]. Since Hulûsi Behçet first postulated a viral etiology for BD in 1937 [5], many studies have suggested a viral involvement in the disease [6,7]. By DNA–DNA dot blotting and polymerase chain reaction, herpes simplex virus (HSV) DNA was also detected in patients' leukocytes and saliva [8,9]. However, treatment with acyclovir, which is of proven efficacy in the treatment of HSV infection, did not significantly alleviate the frequency or severity of orogenital ulceration or other clinical features of BD [10]. Further studies by several research groups have suggested that immune cell dysfunction triggered by HSV might be related to the pathogenesis of this disease, and has been reported by several research groups [11–14]. Until now, the drugs used for the treatment of BD have not been totally effective in controlling symptoms. As part of the search for other agents, which may be effective in the treatment of BD symptoms, we applied the herbal medicine Herba Taraxaci combined with colchicine, a drug that is widely used as a medication for BD. Herba Taraxaci (*Tarazacum mongolicum* Hand.-Mazz.) is derived from the dried whole plant and consists of the taraxasterol, choline, inulin, and pectin [15]. Its known effects include reduction of heat and swelling and clearance of toxic material at sites of inflammation. Herba Taraxaci has been used in the treatment of upper respiratory tract inflammation, pneumonia, acute hepatitis, acute cholecystitis, urethritis, and duodenal ulcers [16]. In recent in vitro studies, Herba Taraxaci has been found to have a strong antibiotic effect against *Staphylococcus aureus* and *Streptococcus pyogenes* as well as *Streptococcus pneumoniae* and *Pseudomonas aeruginosa*. It has also been shown to have an antiviral effect against herpes simplex virus and to enhance immunological responses in vitro [17,18].

2. Materials and methods

2.1. Animals and induction of BD symptoms

Four to 5-week-old male ICR mice were used for this study. Using the method of Hirata et al. [19],

the earlobes of the mice were scratched with a needle, then inoculated with 1.0×10^6 plaque forming units/ml of HSV type 1 (F strain). Virus inoculation was performed twice in a 10-day interval, followed by 16 weeks of observation [20]. Mice were bred in temperature- and light-controlled conventional rooms with 20–22 °C, 12-h light cycle starting at 8:00 a.m. The mice had free access to food and water. During the experimental period, the animals were closely observed and photographed. The animals were handled in accordance with the protocol approved by our institutional animal care committee.

2.2. Gross observation of BD-like symptoms

In order to classify the symptomatic mice as having BD, we followed the revised Japanese classification with minor modifications [20]. Oral, genital, and other skin ulcers (including bulla and crust) and eye symptoms were classified as major symptoms. Arthritis, gastrointestinal ulcers, and neurologic involvement were identified as minor symptoms. Mice with at least one major and one minor symptom were classified as having BD-like symptoms. Symptomatic mice were photographed with a Nikon FM2 camera equipped with a 105-mm micro-lens.

2.3. Dosages and duration of colchicine and Herba Taraxaci treatment

	Dosage	Duration
colchicine	2 µg/day/mouse	20 days
Herba Taraxaci	6 mg/day/mouse	20 days
colchicine + Herba Taraxaci	2 µg/day/mouse + 6 mg/day/mouse	20 days

2.4. Extraction of herbal medicine

Herba Taraxaci was boiled in distilled water of three times of the weight of herb for 3 h and filtered with gauze, then lyophilised until filtered solution change to powder. The whole plant was boiled, filtered, and lyophilised, then 10% of the weight was remained. Six milligrams was re-solved in 200 µl distilled water and orally administered to one mouse per day. The

voucher specimens (ACB 02002) are deposited at the herbarium in the Ajou University.

2.5. Study protocol

Following the development of HSV induced BD-like symptoms, mice were orally administered Herba Taraxaci alone or in combination with colchicines once per day for 20 consecutive days. As a control, nontreated BD-like mice were maintained for the same period. The number of mice in each group was 10. After treatment, half of the mice in each group were sacrificed for RT-PCR and FACS analysis. The remainders were maintained for further 10 days in order to study changes in symptoms and recurrence. Cytokine production was assessed by RT-PCR and FACS analysis to determine whether Herba Taraxaci could influence cytokine production. The number of BD-like mice in each group used for cytokine analysis was three.

2.6. RNA isolation and RT-PCR

Total RNA was isolated by acid guanidium thiocyanate–phenol–chloroform extraction [21]. Spleen tissues were homogenized in 1 ml of extraction buffer, composed of 4 M guanidine solution (Aldrich, Milwaukee, WI, USA), 25 mM sodium citrate, pH 7.0, 0.5% sodium *N*-lauroyl sarcosinate (Fisher, Pittsburgh, PA), and 0.1 M 2-mercaptoethanol (Sigma, St. Louis, MO). A 1/10 volume of chloroform/isoamyl alcohol (49:1) was added to the samples, which were incubated on ice for 5 min then centrifuged at $10,000 \times g$ for 15 min at 4 °C. RNA contained in the upper aqueous phase was collected, precipitated with an equal volume of isopropanol, and washed twice in 70% ETOH. RNA pellets were dissolved in distilled water, quantified by OD 260/280 determination and visualized in an ethidium bromide stained agarose gel. Two micrograms of total RNA were reverse-transcribed using a cDNA kit (Gibco BRL, Grand Island, NY, USA), using oligo dT primers and AMV reverse transcriptase to generate cDNA for use as a template in PCR amplifications. Two microliters from the reverse transcriptase reaction were added to 50 μ l reaction mixtures containing 50 mM KCl, pH 8.4, 20 mM Tris–HCl, 2.5 mM MgCl₂, 200 μ M dNTPs, 2.5 U of Taq polymerase (Gibco BRL), and

1.2 μ M primers. Specific primers for cytokines and beta-actin were as follows:

β -actin [22]	sense: 5'-TGG AATCCTGTGGCATCCATGAAAC-3' antisense: 5'-TAAAACGCAGCTCAGTAACAGTCCG-3'
IL-4 [23]	sense: 5'-ACGCCATGCACGGAGATGGAT-3' antisense: 5'-CAAGCATGGAGTTTCC-3'
IL-10 [24]	sense: 5'-AGACTTTCTTTCAAACAAAGGACCAGCTGGA-3' antisense: 5'-CCTGGAGTCCAGCAGACTCAATACACACTGC-3'
IFN- γ [22]	sense: 5'-AGCGGCTGACTGAACTCAGATTGTAG-3' antisense: 5'-GTCACAGTTTTCAGCTGTATAGGG-3'
Perforin [25]	sense: 5'-AGCTGAGAAGACCTATCAGG-3' antisense: 5'-GATAAAGTGCGTGCCATAGG-3'
Fas [26]	sense: 5'-CTCAAGGTACTAATAGCATCTCCGA-3' antisense: 5'-CAGTAGAAGTCTGGTTTGCACTTG-3'

The amplification was processed in a Perkin Elmer Thermo Cycler 900 with an initial 5 min denaturation at 94 °C, followed by 35 cycles of the profile: 94 °C for 30 s; 56 °C for 30 s; and 72 °C for 1 min. The products were subjected to electrophoresis on a 1.8% agarose gel and visualized under UV light.

2.7. Flow cytometric analysis of intracellular cytokines

Before intracellular cytokine staining, splenocytes were freshly isolated. Brefeldin A (5 g/ml) (Sigma) was added for the last 4 h of incubation to accumulate cytokines in the Golgi complex. Cells were harvested, washed in culture medium containing brefeldin A and fixed with 4% formaldehyde in 1% fetal bovine serum containing PBS for 20 min at room temperature. Then, cells were permeabilized with 0.1% saponin in PBS containing 1% fetal bovine serum and 0.1% sodium azide (saponin buffer) for 10 min at room temperature. Cell suspensions were then treated with FITC- or PE-conjugated anti-IL-4, IFN- γ antibody (CalTag, Burlingame, CA) suspended in permeabilized buffer. Samples were analyzed on a flow cytometer FACS Vantage (Becton Dickinson, San Jose, CA) at least 20,000 gated lymphocytes.

2.8. Statistical analysis

The results were expressed as mean $\pm$ S.E.M. of three of independent experiments in each group. The

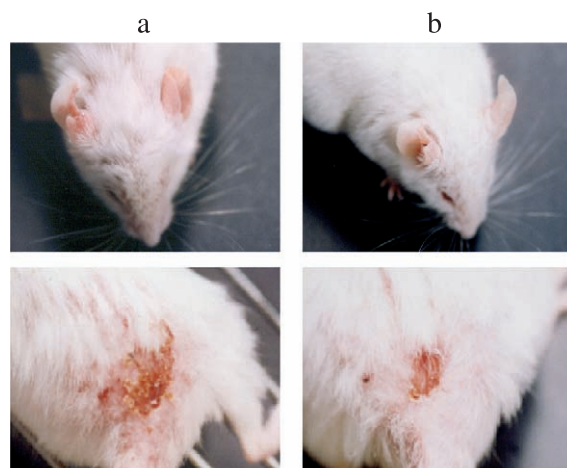


Fig. 1. Improved skin lesions (earlobe, back) of BD-like mouse in response to treatment with colchicine combined with Herba Taraxaci ((a) before treatment, (b) after treatment).

Mann–Whitney test was used for statistical comparisons.

3. Results

A multisystemic disorder with mucocutaneous, ocular, arthritic, vascular, gastrointestinal, and central nervous system involvement appeared in HSV-inoculated mice as previously reported [20], and a mouse with at least one major and one minor symptom was classified as having BD-like symptoms. In order to classify the symptomatic mice as having BD, we followed a revised Japanese classification with minor modifications. Oral, genital and other skin ulcers (including bulla and crust), and eye symptoms were classified as major symptoms, whilst

Table 1

Improvement in symptoms of BD-like mice receiving combined treatment with colchicine and Herba Taraxaci

Treatment	Improved mice/ total mice
Untreated control	0/10 [†]
Colchicine	3/10 [‡]
Herba Taraxaci	6/10 [¶]
Colchicine + Herba Taraxaci	8/10 ^{†, ‡, ¶}

[†] $p < 0.001$.

[‡] $p < 0.025$.

[¶] $p < 0.5$.

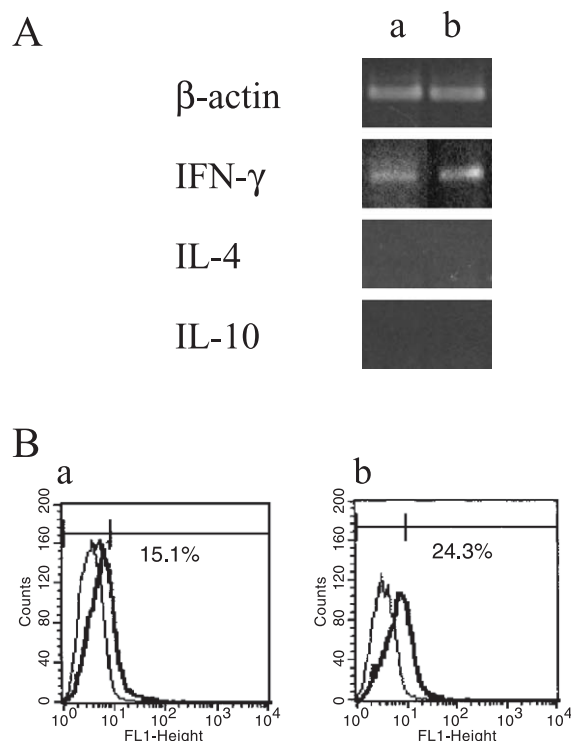


Fig. 2. Cytokine expressions of normal and BD-like mice analyzed by RT-PCR (A) and FACS (B). IFN- γ mRNA was expressed in both normal (a) and BD-like mice (b) ($n = 3$). Protein expression was higher in BD-like mice. The data represent one of three replicate samples from two representative experiments.

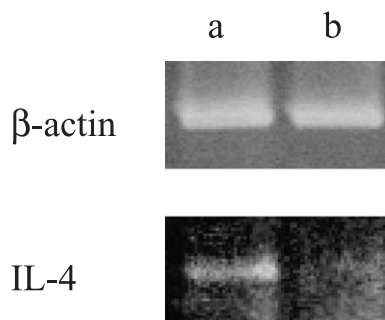


Fig. 3. Cytokine expressions analyzed by RT-PCR in improved BD-like mice treated with Herba Taraxaci (a) or colchicine (b) ($n = 3$). IFN- γ was expressed in both groups, and IL-10 was not expressed in either group. The data represent one of three replicate samples from two representative experiments.

arthritic, vascular, gastrointestinal, and central nervous system involvement were classified as minor symptoms [20].

3.1. *Herba Taraxaci* in mice with BD-like symptoms

BD-like mice treated with colchicine combined with *Herba Taraxaci* showed improvement in symptoms (Fig. 1). Disappearance of symptoms and decrease in lesion size were considered as an improve-

ment. The combined treatment showed the greatest effect with 8 out of 10 mice showing improvement in symptoms compared to 3 out of 10 in the colchicine-only group ($p < 0.025$) and 6 out of 10 treated with *Herba Taraxaci* alone ($p < 0.5$) (Table 1). A range of symptoms was improved by the combined treatment, including skin ulceration in the scruff, peri-ocular skin and earlobes, earlobe erythema, and facial bulla and crust. Large skin ulcers, more than 1.5-cm diameter, were not changed by treatment.

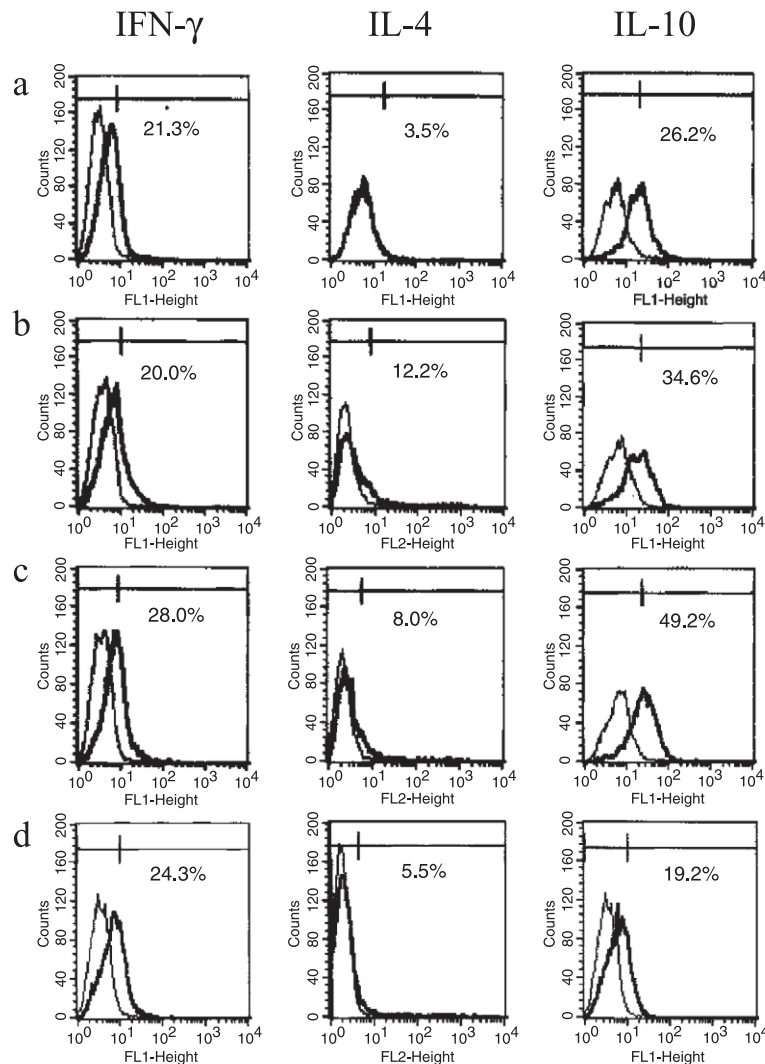


Fig. 4. BD-like mice treated with colchicine and/or *Herba Taraxaci* were analyzed by FACS. (a) colchicine, (b) *Herba Taraxaci*, (c) colchicine and *Herba Taraxaci*, (d) nontreated BD-like mice. The data represent one of three replicate samples from two representative experiments.

Table 2

Percentage of lymphocytes producing IFN- γ , IL-4, and IL-10, as measured by FACS, in treated BD-like mice

Group	IFN- γ	IL-4	IL-10
Colchicine ($n=3$)	20.3 $\pm$ 1.88	3.7 $\pm$ 1.17 [‡]	26.2 $\pm$ 4.47*
Herba Taraxaci ($n=3$)	21.0 $\pm$ 6.31	11.7 $\pm$ 2.56	35.4 $\pm$ 2.17 [¶]
Colchicine + Herba Taraxaci ($n=3$)	26.0 $\pm$ 2.37	8.0 $\pm$ 2.67 ^{†, ‡}	46.8 $\pm$ 6.80*, [¶]
Nontreated ($n=3$)	23.3 $\pm$ 1.98	5.8 $\pm$ 1.45 [†]	18.2 $\pm$ 3.21

Values are the mean $\pm$ S.D.

* $p < 0.001$.

[†] $p < 0.5$.

[‡] $p < 0.25$.

[¶] $p < 0.05$.

After treatment, half of the mice in each group were sacrificed for RT-PCR and FACS analysis. The rest were maintained for a further 10 days in order to observe any further changes in symptoms. During this period, the improvement in symptoms was maintained and BD-like symptoms did not recur.

A side effect of Herba Taraxaci was weight loss, seen as a 16–18% decrease in body weight in normal healthy mice treated with a three times higher concentration (18 mg/day/mouse) for 10 days. However, weight loss was not seen in normal and BD-like mice treated with the concentration used for the study of alleviation of BD-like symptoms (6 mg/day/mouse).

3.2. Effect of Herba Taraxaci on cytokine profiles

Cytokine changes accompanied changes in symptoms. Cytofluorimetric analysis of IFN- γ , a Th1 cytokine, in normal and BD-like mice were demonstrated an increased production in BD-like mice (mean $\pm$ S.D., 23.3 $\pm$ 0.29%) compared with normal mice (14.7 $\pm$ 0.27%) (Fig. 2) ($p < 0.25$). Herba Taraxaci induced IL-4 mRNA in spleen tissues of the BD-like mice as measured by RT-PCR (Fig. 3). Furthermore, FACS analysis showed an increase in the level of IL-4 and IL-10 protein compared to nontreated BD-like mice by FACS analysis (Fig. 4). The combined treatment (colchicine and Herba Taraxaci) group showed the greatest increase in the level of IL-10 protein level (49.2 $\pm$ 6.80%) compared to nontreated BD-like mice (19.2 $\pm$ 3.21%) ($p < 0.001$). IL-4 was also up-regulated by 12.2 $\pm$ 2.56% in splenocytes from mice treated with Herba Taraxaci alone ($p < 0.1$) and 8.0 $\pm$ 2.67% in

the combined treatment group ($p < 0.5$) when compared to nontreated BD-like mice (5.5 $\pm$ 1.45%) (Table 2). In contrast, IFN- γ , Th1 cytokine, production was not significantly changed by either treatment regimen, as shown in Fig. 4. The combined treatment induced symptomatic improvement in BD-like mice, which was accompanied by up-regulation of Th2 cytokine expression compared to nontreated BD-like mice. The onset of symptomatic improvement was earlier with combined treatment of colchicine and Herba Taraxaci treatment (3–20 days), compared to colchicine or Herba Taraxaci alone (6–25 days). Symptomatic improvement induced by colchicine and/or Herba Taraxaci was limited to mucocutaneous manifestations such as crust, erythema, and ulceration. Eye involvement, one of the BD-like symptoms seen in this mouse model, was not affected by treatment with colchicine and/or Herba Taraxaci.

3.3. In vitro effects of Herba Taraxaci on splenocytes cultures

To explore the possible effects of Herba Taraxaci on the regulation of the immune system, in vitro studies were performed on splenocytes cultures. Normal splenocytes were cultured with 0.3 mg/ml Herba Taraxaci then harvested and their mRNA extracted. Results of measurements for mRNA encoding the cell death-related proteins perforin and Fas are shown in Fig. 5. Herba Taraxaci induced both perforin and Fas mRNA at 24 h, but these mRNAs were not longer detectable after 48 h. Splenocytes cultured without

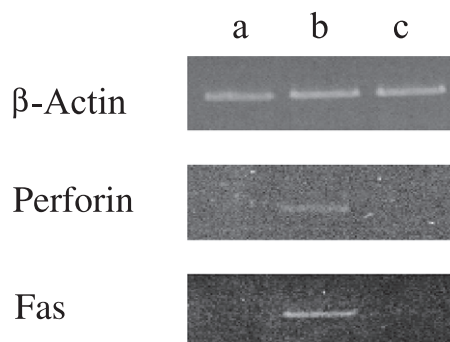


Fig. 5. In vitro induction of cell death-related marker mRNAs by Herba Taraxaci in normal splenocytes culture. (a) 12 h, (b) 24 h, (c) 48 h culture.

Herba Taraxaci showed no detectable perforin or Fas mRNA.

4. Discussion

Herba Taraxaci is an effective and frequently used drug for bacterial and viral infections such as acute respiratory infection, pyogenic skin infections, masitis, jaundice, etc. [18]. Colchicine is also an effective and frequently used drug for BD patients but only produces a limited response. Azathioprine, indomethacin, cyclophosphamide, chlorambucil, levamisole, transfer factor, fibrinolytic therapy, dapsone, and systemic corticosteroids also show limited efficacy in the treatment of this chronic inflammatory disorder. Recently, cyclosporine, thalidomide, interferon- α , interferon- γ , acyclovir, and FK-506 have been trialed, but these drugs did not show any significant difference in efficacy compared to the previously trialed compounds mentioned above [27]. Acyclovir, an antiviral agent, has been tried as a therapeutic drug for BD. However, it showed only negative results as it failed to alleviate the frequency and severity of orogenital ulceration and ocular manifestation [10,28]. Treatment of HSV-1 infected mice with Famciclovir has shown decreased mortality and prolonged the survival time in HSV-1 infected mice [29,30]. However, administration of Famciclovir to HSV-induced BD-like mice reduced symptoms in only 40% of these animals [31]. In the current study, colchicine in combination with Herba Taraxaci reduced in 80% of BD-like mice. Thus, the efficacy of this treatment regimen is apparently greater than that of Famciclovir (40% response) or colchicine (30% response) alone. In this study, Herba Taraxaci was shown to induce IL-4 mRNA, one of the Th2 cytokines that has been associated with improvement in inflammatory diseases. Herba Taraxaci alone or combined with colchicine also increased the number of splenocytes staining with anti-IL-4 antibodies compared to nontreated or colchicine-only treated mice. Herba Taraxaci alone or combined with colchicine also increased the population of splenocytes staining with anti-IL-10 antibodies compared to nontreated or colchicine-only treated mice. According to Noben-Trauth et al. [32], IL-4 drives the differentiation of naïve CD4 positive cells in a Th2 direction. Thus, up-regulated IL-4 mRNA could play a role in delaying the onset of

recurrent BD-like ulceration, in response to HSV, by altering the Th1/Th2 balance in the direction of Th2 cells [35]. Combined treatment with colchicine and Herba Taraxaci also increased the number of IL-4 and IL-10 positive splenocytes compare to nontreated or colchicine treated mice. In work by Bogdan et al. [33], IL-10 has been shown to extensively down-regulate peripheral blood monocytes and macrophages, and inhibit proinflammatory cytokines, and also serve to limit the release of proinflammatory cytokine [34]. Down regulation of macrophages has been shown to decrease the incidence of BD-like symptoms [35]. Increased IL-10 levels induced by colchicine alone would be expected to improve the BD-like symptoms, but the response may be slower than that seen with the combination of colchicines plus Herba Taraxaci. Indeed, the onset of symptom improvement was earlier in the combined treatment group in the current study. The combined treatment group also showed no recurrence of BD-like symptoms for 10 days following the initial improvement.

In vitro treatment of cultured splenocytes with Herba Taraxaci induced perforin, and Fas mRNA were induced at 24 h. Nontreated or colchicine-treated cultures did not express perforin or Fas. The Fas antigen is a cell surface protein belonging to the tumor necrosis factor family. Interaction of Fas antigen with its specific ligand triggers the apoptotic death of T and B lymphocytes [36]. Work by Nakamura et al. [37] suggested that in BD patients with active uveoretinitis, activated CD4 positive T cells with insufficient expression of Fas that would not undergo apoptosis may be responsible for the severe chronic inflammation. Herba Taraxaci might play a role in inducing Fas-triggered cell death of the abnormally proliferated lymphocytes involved in the induction of BD-like symptoms. Perforin, which is normally secreted from cytotoxic T-lymphocytes and natural killer (NK) cells upon conjugation between effector and target cells, is able to insert into the membrane of the target cell. Once there, it then polymerizes to form a cell death-inducing pore through which toxic granzymes may enter the cell and trigger apoptosis [39]. Perforin deficiency is associated with reduced immunopathology [38]. In the one study of human Behcet's disease, perforin was expressed in T cell lines from pustular lesions of the two of four patients with a streptococcal antigen,

KTH-1 [40]. A search of the literature failed to find any other reports regarding the role of perforin in Behcet's disease. Given the above, it is conceivable that increased perforin expression induced by Herba Taraxaci could result in apoptosis of lymphocytes, particularly abnormally proliferated lymphocytes, and hence lead to reduction in inflammation related symptoms.

In conclusion, combined treatment with colchicine and Herba Taraxaci was more effective in improving BD-like symptoms than treatment with either agent alone in an HSV-induced BD-like mouse model. The in vivo improvement was accompanied by increased splenocyte expression of cytokines, IL-4, and IL-10.

Acknowledgements

This work was supported by a grant (99-005-F-00011) from Korea Research Foundation (KRF). We thank Mr. Kwon, H.J., Laboratory of Cell Biology, Ajou University Institute for Medical Sciences, for helpful assistance for animal maintenance.

References

- [1] Namba K, Ueno T, Okita M. Behcet's disease and streptococcal infection. *Jpn J Ophthalmol* 1986;30:385–401.
- [2] Hori Y, Miyazawa S, Nishiyama S, Miyata M, Ishikawa S. Experimental Behcet's disease and ultrastructural X-ray microanalysis of pathological tissues. *J Dermatol* 1979;6:31–7.
- [3] Lehner T. Behcet's syndrome and autoimmunity. *Br Med J* 1967;1:465–7.
- [4] Ohno S, Nakayama E, Sugiura S, Itakura K, Aoki K. Specific histocompatibility antigens associated with Behcet's disease. *Am J Ophthalmol* 1975;80:636–41.
- [5] Behcet H. Über rezidivierende, aphthöse durch ein Virus verursachte Geschwüre am Mund, am Auge und an den Genitalen. *Dermatol Wochenschr* 1937;105:1152–7.
- [6] Sezer FN. The isolation of a virus as the cause of Behcet's disease. *Am J Ophthalmol* 1953;36:301–15.
- [7] Eglin RP, Lehner T, Subak-Sharpe JH. Detection of RNA complementary to herpes-simplex virus in mononuclear cells from patients with Behcet's syndrome and recurrent oral ulcers. *Lancet* 1982;2:1356–61.
- [8] Lee S, Bang D, Cho YH, Lee ES, Sohn S. Polymerase chain reaction reveals herpes simplex virus DNA in saliva of patient with Behcet's disease. *Arch Dermatol Res* 1996;288:179–83.
- [9] Studd M, McCance DJ, Lehner T. Detection of HSV-1 DNA in patients with Behcet's syndrome and in patients with recurrent oral ulcers by the polymerase chain reaction. *J Med Microbiol* 1991;34:39–43.
- [10] Davies UM, Palmer RG, Denman AM. Treatment with acyclovir does not affect orogenital ulcers in Behcet's syndrome: a randomized double-blind trial. *Br J Rheumatol* 1988;27:300–2.
- [11] Hirohata S, Hashimoto T. Abnormal T cell responses to bacterial superantigens in Behcet's disease. *Clin Exp Immunol* 1998;112:317–24.
- [12] Sugi-Ikai N, Nakazawa M, Nakamura S, Ohno S, Minami M. Increased frequencies of interleukin-2- and interferon-gamma-producing T cells in patients with active Behcet's disease. *IOVS* 1998;39:996–1004.
- [13] Turan B, Gallati H, Erdi H, Gurler A, Michel BA, Villiger PM. Systemic levels of the T cell regulatory cytokines IL-10 and IL-12 in Behcet's disease; soluble TNFR-75 as a biological marker of disease activity. *J Rheumatol* 1997;24:128–32.
- [14] Raziuddin S, Al-Dalaan A, Bahabri S, Siraj AK, Al-Sedairy S. Divergent cytokine production profile in Behcet's disease. Altered Th1/Th2 cell cytokine pattern. *J Rheumatol* 1998;25:329–33.
- [15] National Bonchohak Professors, Taraxaci Herba. In: Kim H, editor. *Bonchohak*, Seoul: Jibmoondang; 1991. p. 155–6.
- [16] Ou M. Chinese–English manual of common used in traditional Chinese medicine. Hong Kong: Joint Publishing; 1989. p. 649–50.
- [17] Huang T. A handbook of the composition and pharmacology of common Chinese drugs. Beijing: Chinese Medicine and Pharmacology Press; 1994. p. 1713.
- [18] Kim HS. Pharmacology of Korean herbal medicine. Seoul: Jipmundang; 2001. p. 155–6.
- [19] Hirata Y, Sugita T, Gyo K, Yanagihara N. Experimental vestibular neuritis induced by herpes simplex virus. *Acta Otolaryngol Suppl* 1993;503:79.
- [20] Sohn S, Lee ES, Bang D, Lee S. Behcet's disease-like symptoms induced by herpes simplex virus in ICR mice. *Eur J Dermatol* 1998;8:21–3.
- [21] Chromczynski P, Sacchi N. Single step methods of RNA isolation by acid guanidium thiocyanate–phenol–chloroform extraction. *Anal Biochem* 1987;162:156–9.
- [22] Murray LJ, Lee R, Martens C. In vivo cytokine gene expression in T cell subsets of the autoimmune MRL/Mp-lpr/lpr mouse. *Eur J Immunol* 1990;20:163–70.
- [23] Chan SY, DeBruyne LA, Goodman RE, Eichwald EJ, Bishop DK. In vivo depletion of CD8+ T cells results in Th2 cytokine production and alternate mechanism of allograft rejection. *Transplantation* 1995;59:1155–61.
- [24] Moore KW, Vieira P, Fiorentino DF, Trounstein ML, Khan TA, Mosmann TR. Homology of cytokine synthesis inhibitory factor (IL-10) to the Epstein-Barr gene BCRF1. *Science* 1990;248:1230.
- [25] Kwon BS, Wakulchik M, Liu CC, Persechini PM, Trapani JA, Haq AK, et al. The structure of the mouse lymphocyte pore-forming protein perforin. *Biochem Biophys Res Commun* 1989;158:1–10.
- [26] Watanabe-Fukunaga R, Brannan CI, Itoh N, Yonehara S, Copeland NG, Jenkins NA, et al. The cDNA structure, ex-

- pression, and chromosomal assignment of the mouse Fas antigen. *J Immunol* 1992;148:1274–9.
- [27] Bang D. Treatment of Behcet's disease. *Yonsei Med J* 1997;38:401–10.
- [28] Bonnet M, Ouzan D, Trepo CH. Plasma exchanges and acyclovir in Behcet's disease. *J Fr Ophtalmol* 1986;9:15–22.
- [29] Weinberg A, Bate BJ, Masters HB, Schneider SA, Clark JC, Wren CG, et al. In vitro activities of penciclovir and acyclovir against herpes simplex virus types 1 and 2. *Antimicrob Agents Chemother* 1992;36:2037–8.
- [30] Wutzler P, Ulbricht A, Farber I. Antiviral efficacies of famciclovir, valaciclovir, and brivudin in disseminated herpes simplex virus type 1 infection in mice. *Intervirology* 1997;40:15–21.
- [31] Sohn S, Bang D, Lee ES, Kwon HJ, Lee SI, Lee S. Experimental studies on the antiviral agent famciclovir in Behcet's disease-like symptoms of ICR mice. *Br J Dermatol* 2001;145:799–804.
- [32] Noben-Trauth N, Hu-Li J, Paul WE. IL-4 secreted from individual naive CD4⁺ T cells acts in an autocrine manner to induce Th2 differentiation. *Eur J Immunol* 2002;32:1428–33.
- [33] Bogdan C, Vodovotz Y, Nathan C. Macrophage deactivation by interleukin 10. *J Exp Med* 1991;174:1549–55.
- [34] Moore KW, O'Garra A, de Waal Malefyt R, Vieira P, Mosmann TR. Interleukin-10. *Annu Rev Immunol* 1993;11:165–90.
- [35] Sohn S, Lee ES, Kwon HJ, Lee SI, Bang D, Lee SJ. Expression of Th2 cytokines decreased the development and improved the Behcet's disease-like symptoms induced by herpes simplex virus in mice. *J Infect Dis* 2001;183:1180–6.
- [36] Ayaslioglu E, Turgay M, Duzgun N, Duman M. Expression of Fas antigen (CD95) on peripheral blood lymphocytes in Behcet's disease. *Clin Exp Rheumatol* 2000;18:538–40.
- [37] Nakamura S, Sugita M, Matoba H, Tanaka S, Isoda F, Ohno S. Insufficient expression of Fas antigen on helper T cells in Behcet's disease. *Br J Ophthalmol* 1996;80:174–6.
- [38] Joseph A, Trapani A, Smyth MJ. Functional significance of the perforin/granzyme cell death pathway. *Nat Rev, Immunol* 2002;2:735–47.
- [39] Henter JI. Biology and treatment of familial hemophagocytic lymphohistiocytosis: importance of perforin in lymphocyte-mediated cytotoxicity and triggering of apoptosis. *Med Pediatr Oncol* 2002;38:305–9.
- [40] Mochizuki M, Morita E, Yamamoto S, Yamana S. Characteristics of T cell lines established from skin lesions of Behcet's disease. *J Dermatol Sci* 1997;15:9–13.